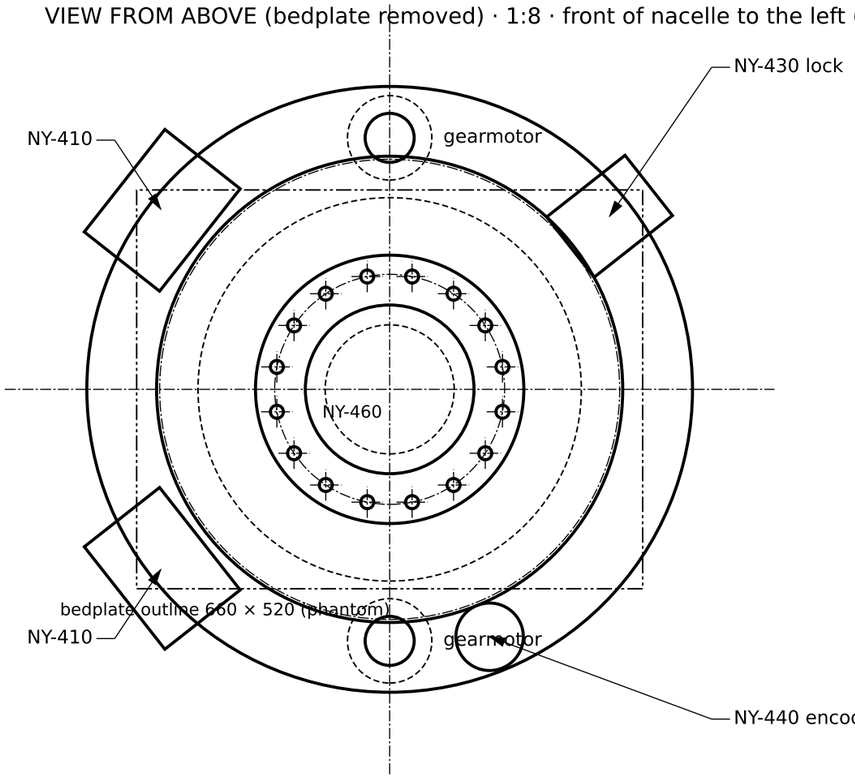
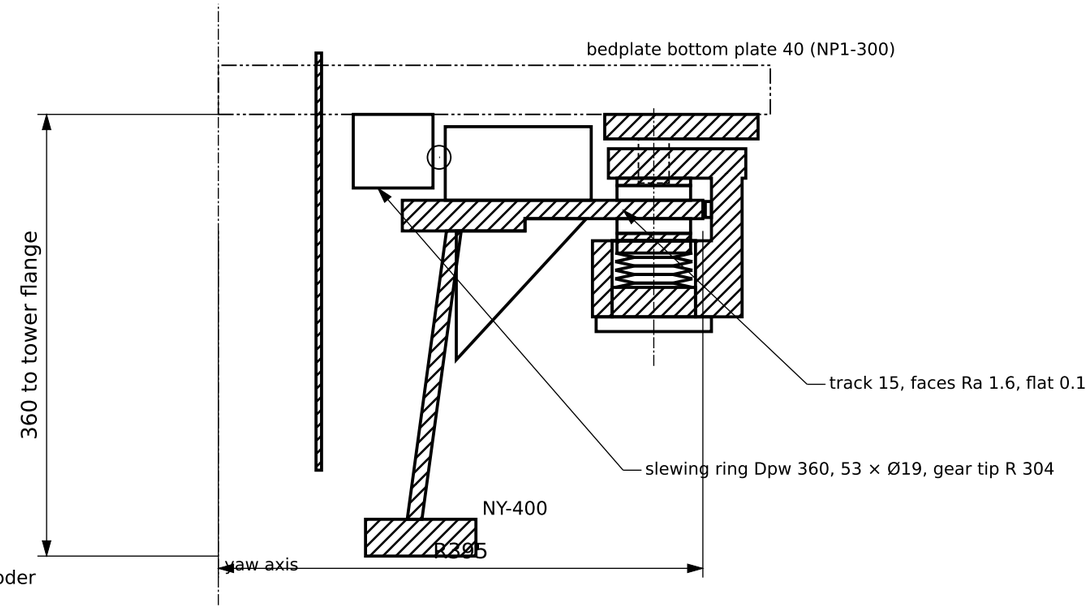
VIEW FROM ABOVE (bedplate removed) · 1:8 · front of nacelle to the left (−X)



SECTION A-A through the pad unit at 141.8° · 1:5



Item	Part	Qty	Make/buy	Mass
1	NY-400 tower-top adapter (S355J2+N weldment)	1	make	107 kg
2	Slewing ring Dpw 360, ext. gear z150 m4, ind. hardened teeth	1	buy	≈ 55 kg
3	Gearmotor 0.18 kW 4-pole, 1:1,680 planetary, 2 Nm motor brake	2	buy	≈ 22 kg
4	NP1-420 drive bracket (doc 16)	2	make	7.7 kg
5	NY-410 friction pad unit	2	make	13 kg
6	NY-430 yaw lock	1	make	6.5 kg
7	NY-440 encoder + cam limit switch unit	1	make/buy	6.9 kg
8	NY-460 cable hanger	1	make	10 kg
9	Bolts: 16 M16 inner, 16 M16 outer (from below), 16 M20 tower	48	buy	
10	Heated 2D ultrasonic + heated cup/vane on the nacelle mast	1+1	buy	

NOTES

1. No hydraulics on the yaw system. Holding: 2 spring-applied motor brakes (fail-safe) through the gear train; damping: 2 passive pad units, 2.4 to 3.6 kNm.
2. Design yaw moment held 8 kNm by ONE pinion (backlash) without friction credit; gearbox static rating ≥ 1,500 Nm output.
3. All bolts 10.9, preload with calibrated torque, M16 90 kN, M20 110 kN. Grease the ring per supplier (−40 °C synthetic).
4. Yaw rate 0.5 °/s at 1,500 rpm. One VFD drives both motors; the safety chain opens the brake supply directly.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

YAW SYSTEM · GENERAL ARRANGEMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NY-000	A · 2026-11-05	1:5 / 1:8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see parts list	≈ 275 kg (yaw incl. adapter)	Claude / not checked